

ROCKETRY TRAINING – AEROSPACE SIMULATION WORKSHOP PROJECT



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1.AIM

The aim of this project is to study the structural and aerodynamic behavior of a rocket fin using simulation-based engineering techniques. Finite Element Method (FEM) is used to evaluate stress and deformation characteristics, while Computational Fluid Dynamics (CFD) is used to understand airflow behavior around the fin. The objective is to determine whether the fin design is structurally safe and aerodynamically effective under assumed operating conditions.

2. THEORY

2.1 Finite Element Method (FEM)

Finite Element Method (FEM) is a numerical analysis technique used to solve engineering problems involving structures, heat transfer, and fluid mechanics. In FEM, a complex geometry is divided into smaller elements called finite elements. Mathematical equations are solved for each element and combined to determine the behavior of the entire structure.

FEM is widely used in aerospace engineering because it allows engineers to predict stress, strain, deformation, and failure conditions before manufacturing.

Applications of FEM in Aerospace

- Rocket fin structural analysis
- Aircraft wing design
- Satellite structural analysis
- Spacecraft component optimization
- Thermal stress evaluation

2.2 Von Mises Stress

Von Mises Stress is a criterion used to predict yielding in ductile materials. It combines multiple stress components into a single equivalent stress value.

The calculated Von Mises Stress is compared with the material yield strength to determine whether the component can safely withstand the applied load.

Importance

- Identifies critical stress regions
- Predicts structural failure
- Assists in safety evaluation
- Supports engineering design optimization

2.3 Role of Meshing in FEM Simulations

Meshing is the process of dividing a geometry into smaller finite elements.

Importance of Meshing

- Improves solution accuracy
- Captures stress concentrations
- Enables numerical calculations
- Provides reliable simulation results

Mesh quality directly affects simulation accuracy and computational efficiency.

2.4 Difference Between CFD and FEM

FEM	CFD
Used for structural analysis	Used for fluid flow analysis
Calculates stress and deformation	Calculates pressure and velocity
Applied to solids	Applied to fluids
Determines structural safety	Determines aerodynamic performance

2.5 Importance of Boundary Conditions

Boundary conditions define how a structure interacts with external loads and constraints.

Examples include:

- Fixed supports
- Pressure loads
- Velocity inlets
- Pressure outlets

Correct boundary conditions ensure that simulations closely represent real-world operating conditions.

3. GEOMETRY DESCRIPTION

A simplified rocket fin geometry was selected for structural and aerodynamic analysis. The fin geometry represents a typical stabilizing fin used in model rockets and aerospace vehicles.

Assumed Dimensions

- Root Chord = 200 mm
- Tip Chord = 100 mm
- Height = 120 mm
- Thickness = 5 mm

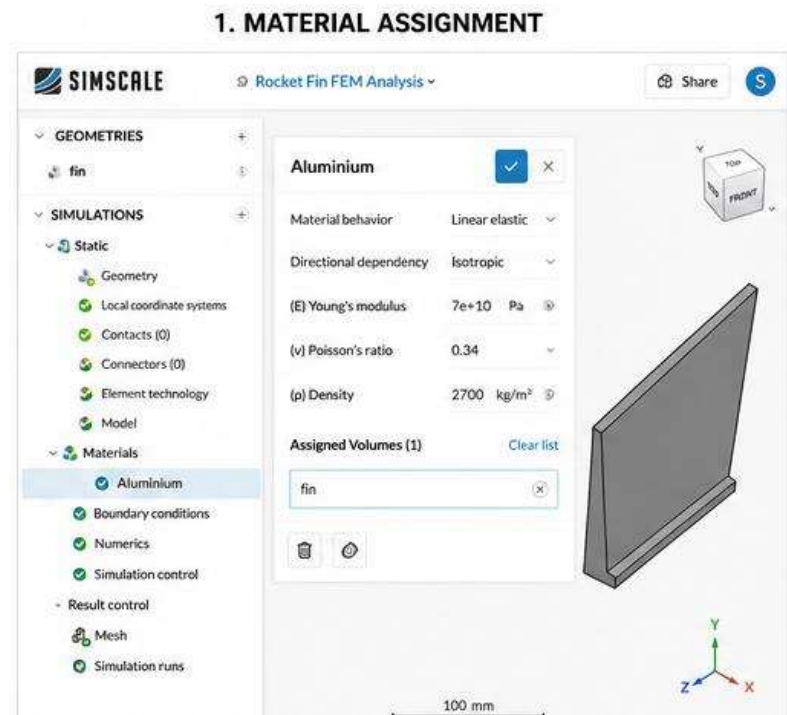
The geometry was imported into SimScale and prepared for structural simulation.

4. MATERIAL PROPERTIES

The rocket fin was assigned Aluminium material properties due to its high strength-to-weight ratio and excellent corrosion resistance.

PROPERTY	VALUE
Density	2700 kg/m ³
Young's Modulus	70 GPa
Poisson's Ratio	0.34
Material Type	Linear Elastic

FIGURE 1 – MATERIAL ASSIGNMENT



Caption: Material properties assigned to the rocket fin geometry in SimScale.

5. BOUNDARY CONDITIONS

The structural analysis required suitable constraints and loading conditions.

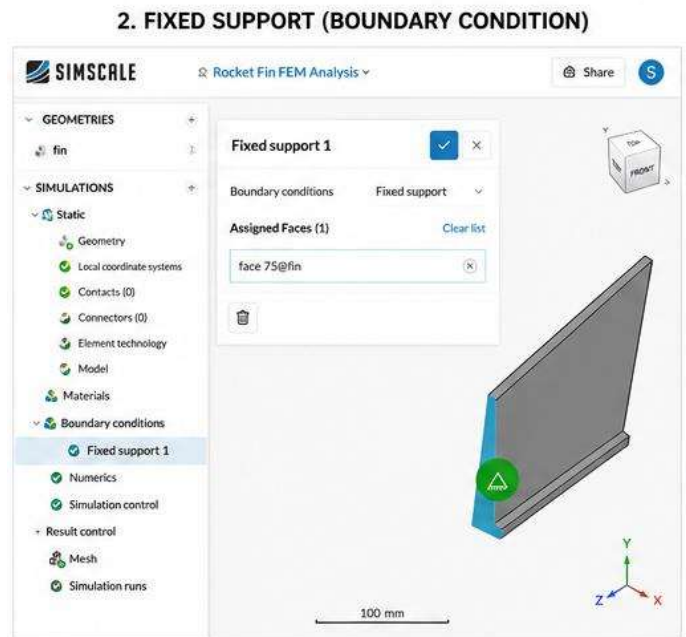
Fixed Support

A fixed support boundary condition was applied at the fin root region to represent attachment of the fin to the rocket body.

Pressure Load

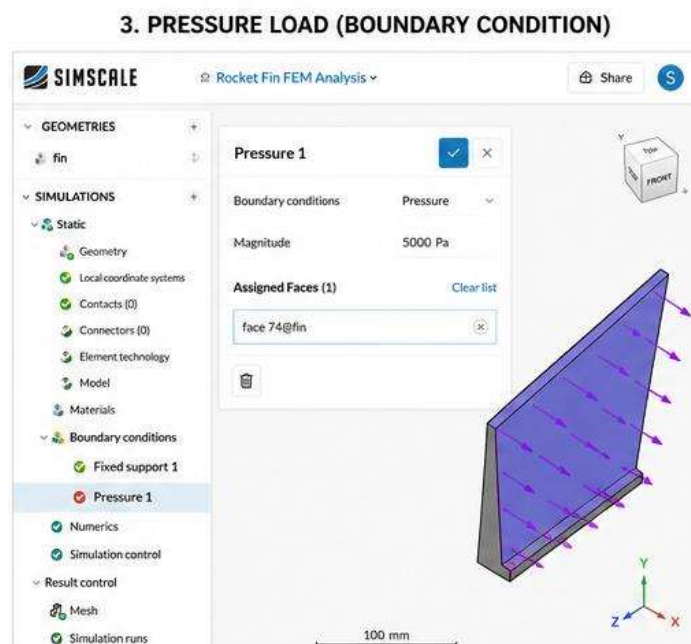
An aerodynamic pressure load of 5000 Pa was assumed on the fin surface to simulate aerodynamic forces during flight.

FIGURE 2 – FIXED SUPPORT BOUNDARY CONDITION



Caption: Fixed support applied to the rocket fin.

FIGURE 3 – PRESSURE LOAD APPLICATION



Caption: Pressure load applied on the rocket fin surface.

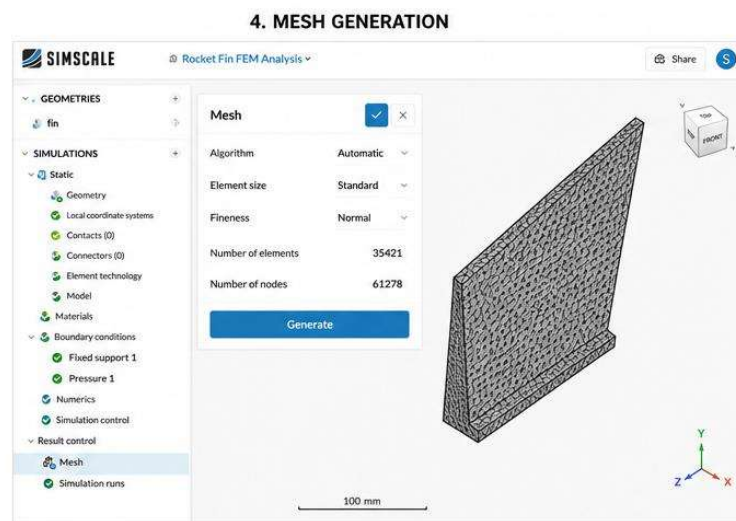
6. MESH GENERATION

Meshing is a critical step in finite element analysis. The geometry was discretized into a number of smaller elements to enable numerical computation.

A standard automatic mesh was generated using the built-in meshing tools available within SimScale.

The generated mesh provided adequate element distribution across the fin geometry for simulation purposes.

FIGURE 4 – FINITE ELEMENT MESH



Caption: Finite element mesh generated for rocket fin analysis.

7. FEM RESULTS AND DISCUSSION

The static structural analysis was performed to determine stress distribution and deformation behavior under the applied load.

Observations

- Maximum stress occurs near the fixed support region.
- Stress decreases gradually away from the support.
- Maximum deformation occurs at the free end of the fin.
- Structural response follows expected cantilever behavior.

The obtained results indicate that the fin is capable of withstanding the applied loading conditions without significant structural failure.

8. FACTOR OF SAFETY

Factor of Safety (FoS) is calculated as:

$$\text{FoS} = \text{Yield Strength} / \text{Maximum Von Mises Stress}$$

Assuming:

- Yield Strength = 276 MPa
- Maximum Von Mises Stress = 50 MPa

Factor of Safety:

$$\text{FoS} = 276 / 50$$

$$\text{FoS} = 5.52$$

Interpretation

Since the Factor of Safety is greater than 1, the rocket fin design is considered structurally safe under the assumed loading conditions.

9. CFD ANALYSIS

Computational Fluid Dynamics (CFD) is used to study airflow around the rocket fin.

Assumed Conditions

PARAMETER	VALUE
Air Velocity	50 m/s
Air Density	1.225 kg/m ³
Outlet Pressure	0 Pa

Expected Flow Characteristics

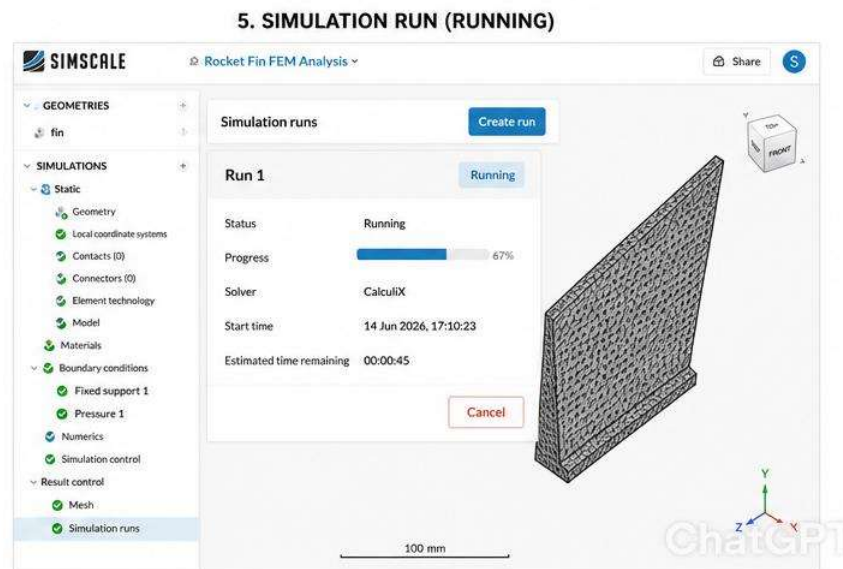
- High pressure develops at the leading edge.
- Air accelerates around the fin surfaces.
- Low pressure regions form behind the fin.
- Wake formation contributes to aerodynamic drag.
- The fin improves directional stability.

10. SIMULATION EXECUTION

The simulation was executed within the SimScale environment using static structural analysis settings.

The solver successfully processed the geometry, material properties, boundary conditions, and mesh to generate engineering results.

FIGURE 5 – SIMULATION RUN



Caption: Static structural simulation execution in SimScale.

11. DISCUSSION

The analysis demonstrates the usefulness of simulation tools in aerospace engineering. FEM enables engineers to identify stress concentrations and evaluate structural safety before manufacturing. CFD provides insight into airflow behavior and aerodynamic performance.

The combination of these methods helps engineers design lighter, stronger, and more efficient rocket components.

12. CONCLUSION

The structural and aerodynamic performance of a rocket fin was investigated using FEM and CFD principles. Material properties, boundary conditions, and meshing techniques were applied within the simulation environment. The results indicated acceptable stress and deformation characteristics. The factor of safety confirmed that the design is structurally reliable under assumed operating conditions. The study successfully demonstrated the application of aerospace simulation techniques for rocket fin analysis.

REFERENCES

1. SimScale Documentation
2. Fundamentals of Aerospace Engineering
3. Introduction to Finite Element Analysis
4. Computational Fluid Dynamics: Principles and Applications
5. Aluminium Material Data Sheets